CH32M030_Motor Evaluation Board Reference

Version: V1.1
https://wch-ic.com

1. Overview

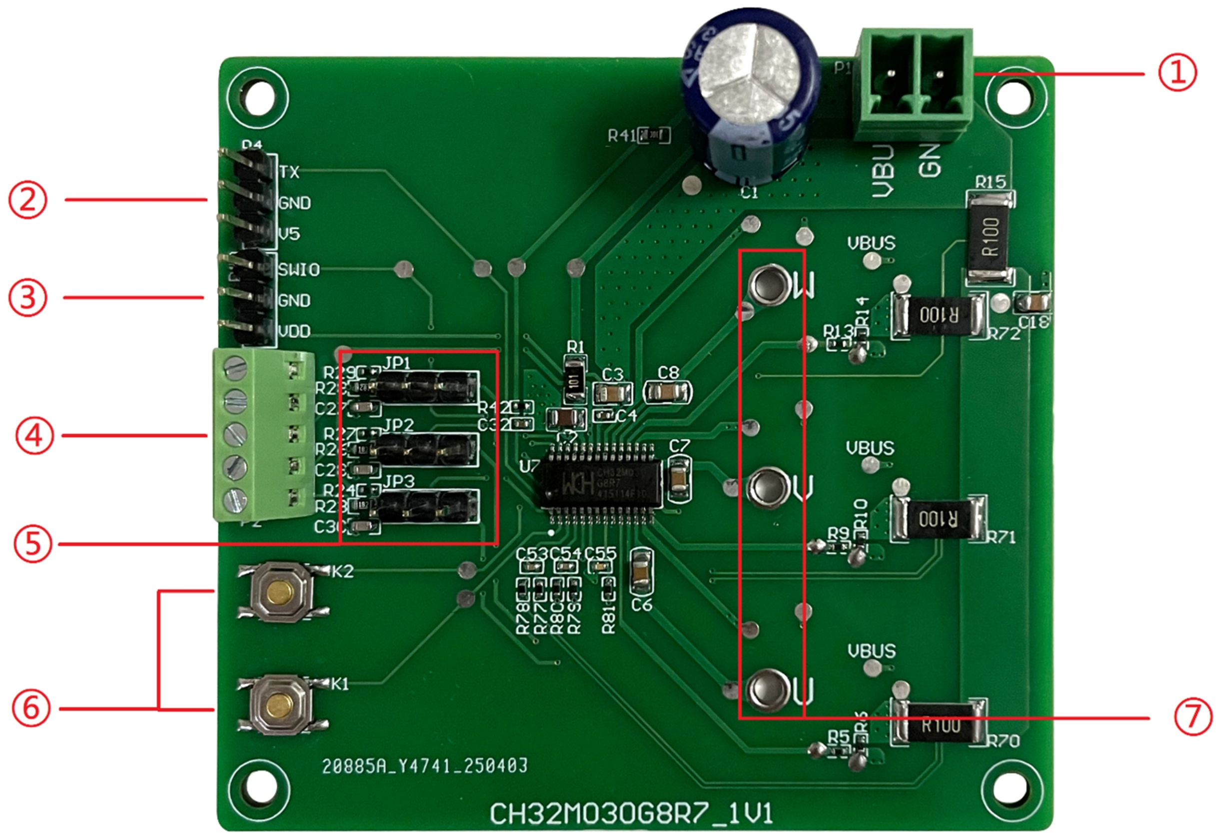
This evaluation board is only suitable for demonstrating motor application routines under the EVT path (EVT->EXAM->APPLICATION-> Electric_Fan). The main control chip is CH32M030G8R7.

This application solution boasts advantages such as low loss, stable operation, and low BOM cost. It can drive a 100,000rpm three-phase brushless ultra-high-speed motor, offering customers a complete and customizable mass production solution.

2. Evaluation Board Hardware

Please refer to the CH32M030SCH_Motor.pdf document for the schematic of the evaluation board.

CH32M030 Evaluation Board



Descriptions

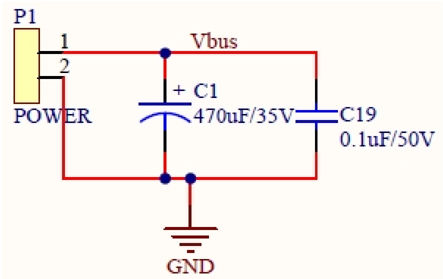
1. Power interface	2. Virtual oscilloscope interface	3. SWD port
4. Hall sensor interface	5. Jumper socket	6. Custom button
7. Motor interface		

Tips: The debug interface of CH32M030 series chips supports free configuration; 1-wire debugging or 2-wire debugging is optional. Debugging interface pins PA3 (SWIO), PA2 (SWCLK 2-wire debugging optional)

2.1 The Above Evaluation Board Consists of the Following Modules

2.1.1 Power Module

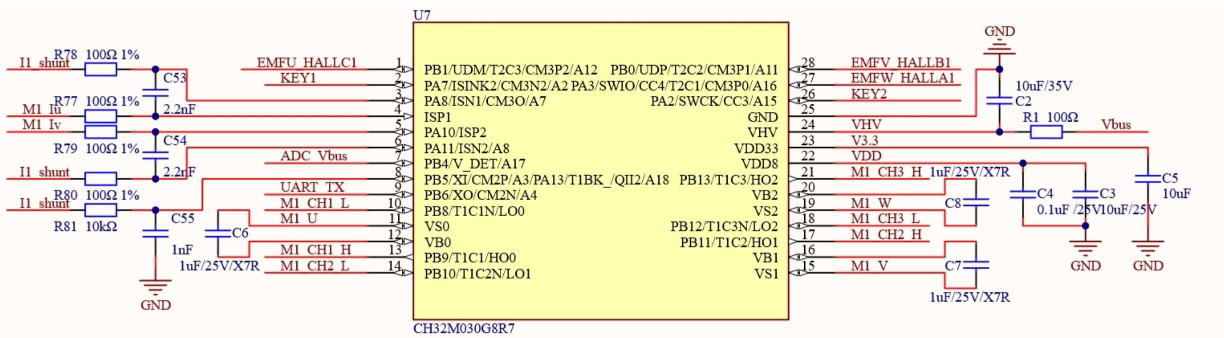
Figure 1 Power input interface



As shown in Figure 1, the input voltage of the power input interface does not exceed 28V.

2.1.2 MCU Circuit

Figure 2 MCU circuit



As shown in Figure 2, the main control MCU is CH32M030G8R7, which is internally provided with double N-type power transistor grid pre-driving, so that the driving circuit can be driven only by three external bootstrap capacitors; Provide high voltage I/O, built-in 4 OPAs and 3 CMPs, and support 2 groups of differential input current sampling ISP and 2 groups of QII; The MCU can complete the functions of single and double resistance current amplification and sampling, overcurrent comparison and back EMF comparison in very few peripheral scenes. Built-in high-side and low-side LDOs provide pre-drive and MCU power supply, so that MCU can work normally without external power supply; Includes a 16-bit advanced-control timer with dead-time control and braking, which is used for PWM complementary output of motor, a general-purpose timer and a streamlined timer.

In Figure 2, C6, C7 and C8 are bootstrap capacitors with internal integrated pre-drive.

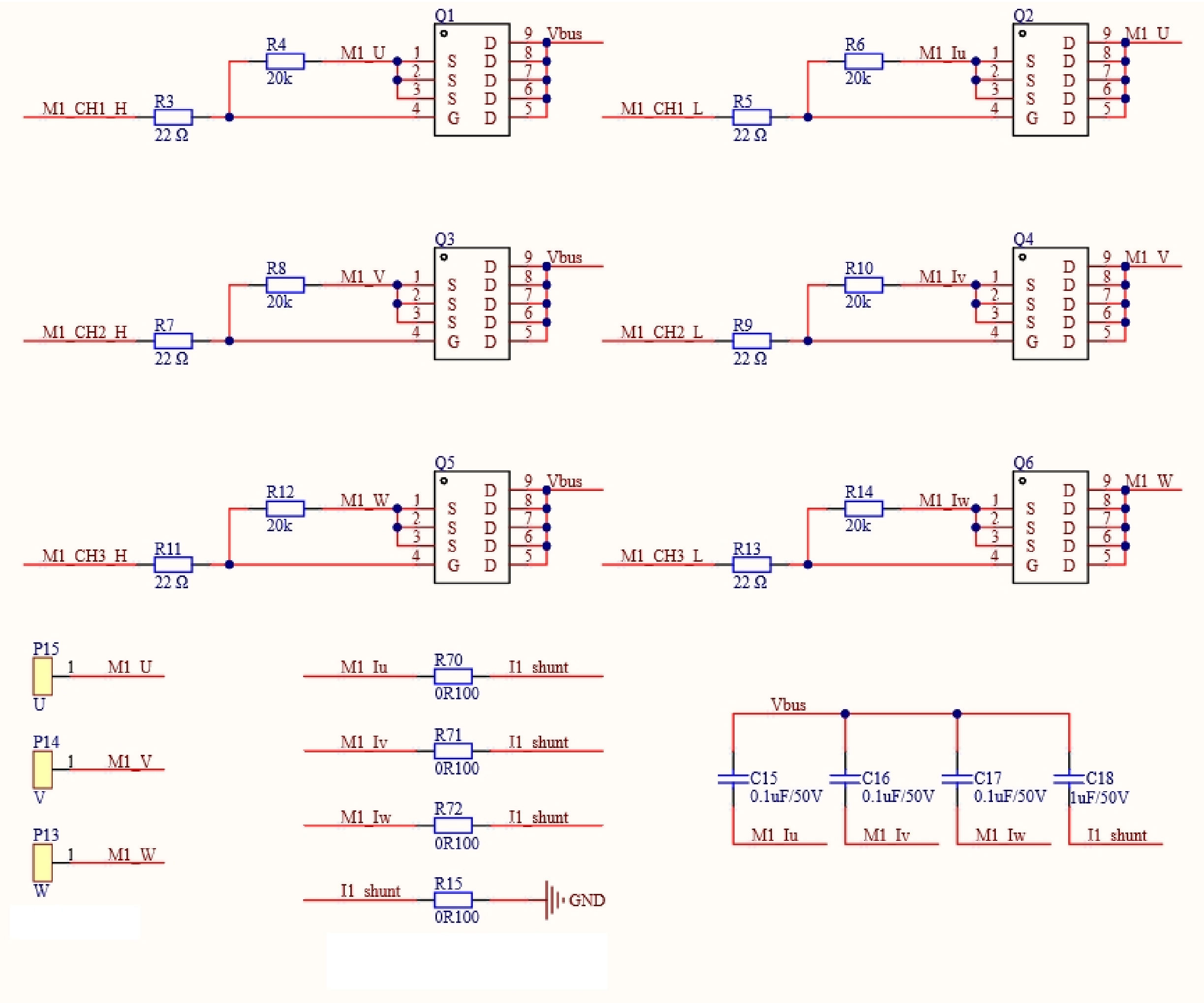
In Figure 2, the bus power supply is connected to the HV pin of MCU through R1, and after passing through LDO, a pre-drive power supply voltage VDD8 can be obtained, which can be configured to be 5~10V through a register, and then the working voltage of VDD33——MCU can be obtained through an LDO. The package size of R1 is determined by the input voltage and current.

In Figure 2, M1_Iu and I1_shunt, M1_Iv and I1_shunt are two-phase current samples of U and V, which are respectively connected to internal OPA3 and OPA4 through differential wiring, and the output results are sent to the internal ADC channel to complete phase current sampling.

In Figure 2, another path of I1_shunt is filtered by R81 and C55, and then sent to the internal QII2. After internal amplification and comparison with the set value of the comparator, the comparison result is sent to the BKIN channel of TIM1 to trigger the bus overcurrent.

2.1.3 Inverter Circuit

Figure 3 Inverter circuit (electric control board)

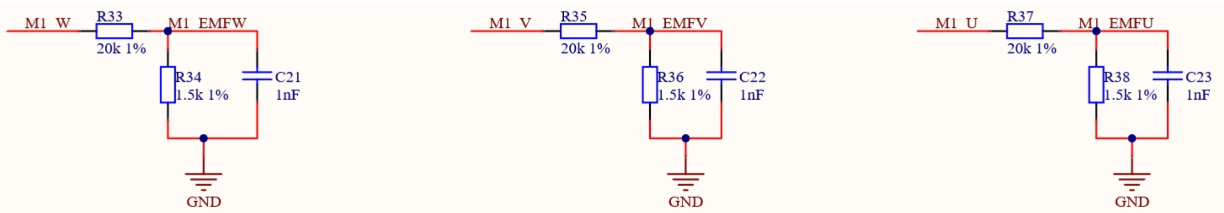


As shown in Figure 3, the inverter circuit consists of six NMOS transistors of type NP160S04PD6, which convert control commands from the MCU into signals for motor operation. R70 and R71 are used for dual-resistor sampling, while R72 primarily balances the internal resistance across the three phases for low-impedance motors; it can be omitted in conventional designs. R15 is employed for bus current sampling. C15, C16, C17, and C18 are typically connected between the upper bridge drain and lower bridge source, positioned adjacent to their corresponding MOS. This arrangement improves control performance and reduces oscillations within the loop.

As shown in Figure 3, P13, P14, and P15 serve as the motor connection interfaces, linking to the motor's W, V, and U phases respectively.

2.1.4 Phase Voltage Sampling Circuit

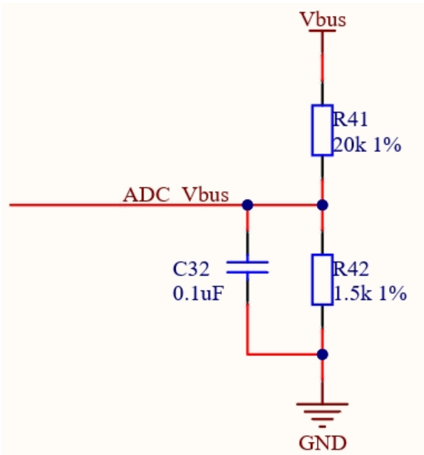
Figure 4 Sampling current of phase voltage



As shown in Figure 4, in sensorless BLDC applications, back EMF is typically required for position detection. This can generally be achieved through back EMF sampling or by comparing the back EMF with a virtual center point. In practical applications, the ratio and resistance value of the resistors are adjusted based on the actual phase voltage.

2.1.5 Bus Voltage Sampling Circuit

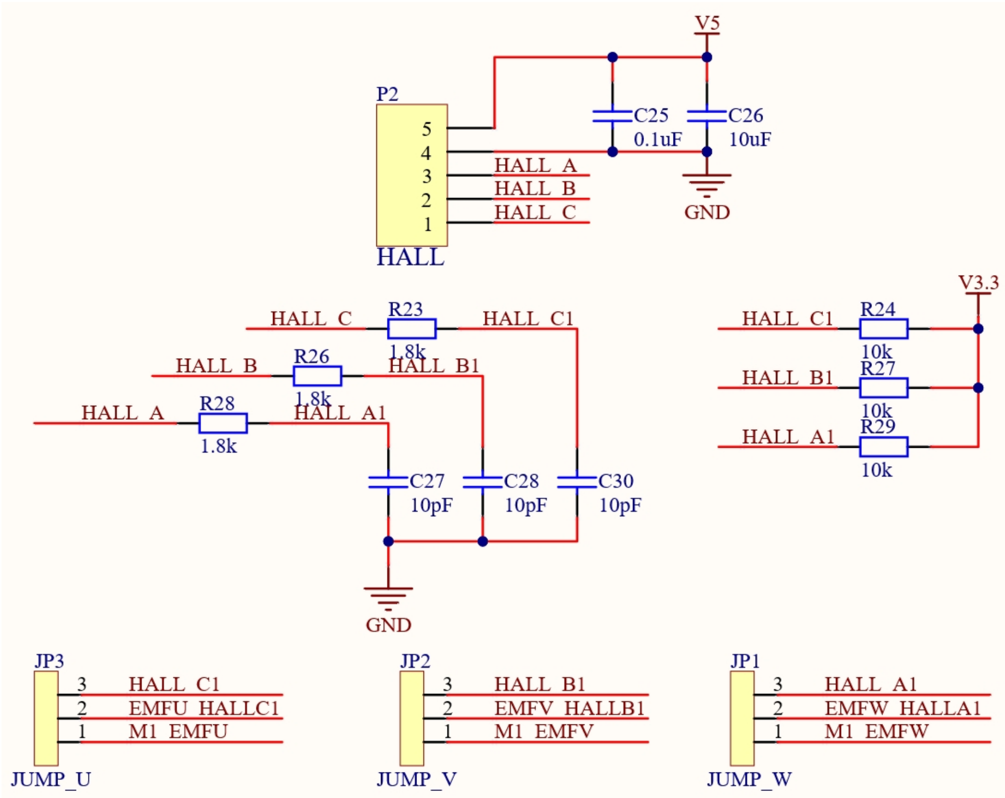
Figure 9 Bus voltage sampling circuit



As shown in Figure 5, the bus voltage is sampled using a voltage divider. In practical applications, the ratio and resistance values of R41 and R42 are adjusted based on Vbus.

2.1.6 Hall Sensor Interface Circuit

Figure 6 Hall sensor interface circuit

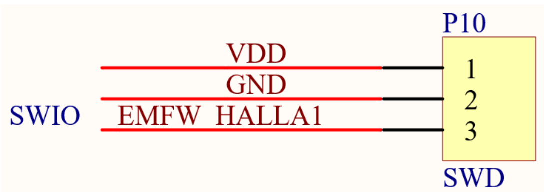


As shown in Figure 6, P2 is the Hall sensor interface. After RC filtering and pull-up, the signal outputs to the MCU via HALL_A1, HALL_B1, and HALL_C1 to the MCU. Since these interfaces are shared with the back-EMF detection mode, jumper settings route the signals via EMFU_HALLA1, EMFV_HALLB1, and EMFW_HALLC1 to the MCU's PA3, PB0, and PB1 pins. These pins can be configured as capture interfaces for General-Purpose Timer 2. The 5V supply is externally connected for dedicated power.

As shown in Figure 6, the JP1, JP2, and JP3 jumper connectors are used to select either the Hall signal or back-EMF signal fed into the MCU, supporting both inductive and non-inductive solutions.

2.1.7 SWD Interface

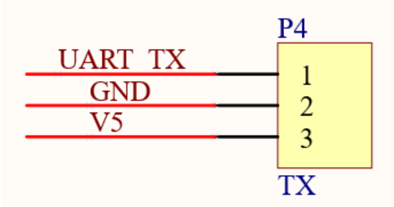
Figure 7 SWD interface



As shown in Figure 7, SWD interface can be used to simulate debugging and downloading. This type of MCU supports 4-wire download (as shown above) and is also compatible with 3-wire mode (power supply +SWIO).

2.1.8 Virtual Oscilloscope Observation Interface

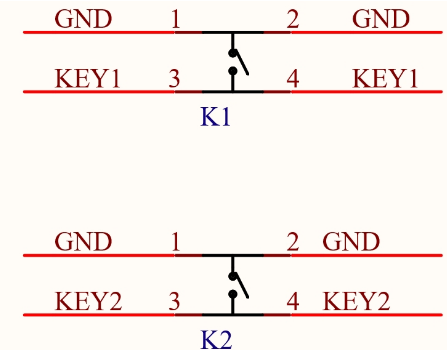
Figure 8 Virtual oscilloscope observation interface



As shown in Figure 8, the serial port is used to connect the adapter board and cooperate with the upper computer to observe and debug. 5V provides separate power supply for external access.

2.1.9 Custom Buttons

Figure 9 Custom buttons



As shown in Figure 9, 2 buttons can be used to customize functions.